



12090 - Chemical Modeling for Ultra-cool Brown Dwarfs and Giant Planets

Cycle: 5, Proposal Category: AR

INVESTIGATORS

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OBSERVATIONS

ABSTRACT

JWST is a phenomenal tool for studying the coldest exoplanets and ultra-cool brown dwarfs. Soon the study of exoplanet and brown dwarf atmospheres with temperatures that overlap those of the solar system planets will be routine. The atmosphere models we use to interpret JWST observations rely on chemical models that form the bridge between the absorption features we measure and the element abundances we seek. Motivated by model-observation discrepancies revealed by JWST spectroscopy, this proposal focuses on improving the modeling of disequilibrium chemistry driven by vertical mixing and height-dependent element abundances established by condensation.

OBSERVING DESCRIPTION